Depth as sampling geometry, and the quorum as a 2×2 mechanism

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1 September 2026; v4, 18 October 2026: corrected cross term and replayed 2×2 certificates

Abstract

Two results replace two earlier readings. (1) The depth law $-\ln \lambda_{\min} = s(\chi)\mu + b$ of the truncated Weil forms, previously described by a four-variable regression on fourteen Dirichlet characters that failed out of sample, is a *geometric* property of the zero set: the floor c_L of the form is the sampling constant of the zeros for the Paley–Wiener space of type $L/2$, made exponentially small by the desert $(-\gamma_1, \gamma_1)$ and by the gaps longer than the Nyquist step $2\pi/L$. A two-coefficient law, the coefficients fitted on four scales of ζ alone ($\mu = 3, 8, 11, 16$), predicts $s(\chi)$ out of sample with median ratio 0.89 (10 of 14 characters within 20%), with no character-specific parameter. (2) The quorum theorem (every proper sub-Euler product violates positivity) has a mechanism: an unconditional 2×2 interlacing lemma, *depth* $\times$ *coupling*, whose three quantities we certify at $\mu = 11$ and measure at three scales; the coupling of the bottom vector to a missing prime decays in p at half the rate of its silence, which is what makes the determinant negative. Both are statements about explicitly defined forms; nothing bears on RH.

Status of Section 1 (5 September 2026). The two-term formula $-\ln c_L \approx aL(\gamma_1 - \nu)_+ + bL\sum(\gamma_{k+1} - \gamma_k - \nu)_+$ is *dead as a quantitative law*. Its gap sum does not converge at accessible heights, so (a, b) depend on the height cutoff of the zero lists, and the out-of-sample table below compared ζ summed to 811 with characters summed to 150. At a common cutoff $T_0 = 320$ (complete one-sided Weyl lists for sixteen L-functions; $(a, b) = (1.71, 0.97)$ from ζ alone) the formula overpredicts most L-functions (ratios 0.57–0.93) and underpredicts χ_{24}^o by 2.2; three conductors measured after the definition $(\chi_{29}, \chi_{17}, \chi_5)$ come out at 0.57–0.70 (notebook §92–94, `report/verdict-T0-320.md`). What survives is the *structure*: the floor is a sampling constant made small by the desert and by the sub-Nyquist gaps, but their combination is not additive; the right object is the sampling constant of the one set E_L = desert $\cup$ long gaps (`report/one-set-sampling.md`), whose value is open. Section 2 is unaffected.

1 Depth is a sampling constant

Everything in this section is under RH (equivalently, under the identification $Q = Q_z$ of the prime-side form with the zero Gram): it describes the floor once the zeros are on the line. Section 2 uses no zeros at all. Under RH, $Q_L(f) = \sum_\gamma \hat{f}(\gamma)^2$ with $\hat{f} \in PW_{L/2}$, so $c_L = \inf Q_L/\|f\|^2$ is a sampling constant of the zero set, positive by Beurling (note *sampling-floor*, Theorem 1). The finite-basis minima have a floor. At $\mu = 3$ the saturation is visible: $\lambda_{\min}(N) = 1.026, 0.775, 0.641, 0.591, 0.571, 0.562 \times 10^{-7}$ for $N = 9 \ldots 61$, successive ratios $1.32, 1.21, 1.08, 1.03, 1.02$, converging geometrically to $c_{\log 3} \approx 5.55 \times 10^{-8}$. At $\mu = 11$ it is so far a trend: $3.59, 1.86, 1.54 \times 10^{-48}$ for $N = 47, 57, 67$, ratios 1.93 then 1.21 — still a slope, decelerating toward something near 1.4×10^{-48}, not yet a saturation.

Synthetic controls (zero-side Gram, $\mu = 3$). $Q_z = \sum_\gamma 2\hat{c}(\gamma)\hat{c}(\gamma)^\top$ reproduces the prime-side floor (ratio 0.93, the truncation tail). Three fictitious zeros placed in the desert raise $\lambda_{\min}$

by 9×10^6; a comb with the same γ_1 and density changes it by 0.23 only; shifting the whole spectrum changes $-\ln \lambda_{\min}$ linearly in the desert width with slope 1.45 against $L = 1.10$ for Slepian's single hole. The desert is the cause; the arithmetic fine structure touches only the prefactor.

The geometric law. With $\nu = 2\pi/L$ the Nyquist step,

$$-\ln c_L \;\approx\; a\,L\,(\gamma_1 - \nu)_+ \;+\; b\,L \sum_k (\gamma_{k+1} - \gamma_k - \nu)_+, \qquad a = 1.69,\ b = 0.82,$$

the two coefficients fitted on the μ-series of ζ only ($\mu = 3, 8, 11, 16$, within 8%; Slepian's naive values would be $2, 1$). Let $\gamma^*(L)$ be the last height at which a consecutive gap exceeds ν (for $\mu = 11$: gaps above $\nu = 2.62$ occur up to $\gamma \approx 300$ in the first 500 zeros, sporadically, while the band edge of V_{47} is $\omega_{\max} = 120.5$). As L grows, ν shrinks and the excess sum extends and grows; over $\mu = 3, 8, 11, 16$ it is empirically linear in μ, which is where the linearity of the depth law comes from in this description.

Out of sample (coefficients frozen). Predicted s = slope of the law between $\mu = 11$ and 22, against the measured s of the depth note:

χ	ζ	χ_3	χ_4	χ_5	χ_7	χ_8	χ_{11}	χ_{12}
ratio	1.22	0.82	0.82	0.87	1.08	1.00	1.17	0.88
χ	χ_{13}	χ_{15}	χ_{19}	χ_{21}	χ^e_{24}	χ^o_{24}	χ_{-23}	
ratio	1.35	1.02	0.96	0.65	1.04	0.38	0.89	

Median 0.89; ten of fourteen within 20%; no parameter per character. The downward bias is mechanical (zero caches end near $\gamma \approx 85 < \gamma^*(22) = 138$). Of the two failures, χ^o_{24} ($\gamma_1 = 1.98$) has $\gamma_1 < \nu$ at both scales ($\nu(11) = 2.62$, $\nu(22) = 2.03$): its desert term is identically zero, and the linearized $(\gamma_1 - \nu)_+$ is the culprit — Slepian's $1 - \lambda_0(c)$ does not switch off. χ_{21} ($\gamma_1 = 2.32$) sits between $\nu(22)$ and $\nu(11)$: the term switches on between the two scales, which explains only part of the 0.65; the rest is an unexplained residue. Where the four-variable map (trained on these characters) died out of sample (χ_{-23}: -29%), the geometric law predicts χ_{-23} to 0.89.

Caveat on the gap term (5 September). $\sum_k (\gamma_{k+1} - \gamma_k - \nu)_+$ does not converge at accessible heights: for ζ at $\mu = 22$ it grows from 38.9 to 70.5 as the zero list is cut at 150 and 811 (the number of gaps above ν rising from 32 to 111; the mean spacing $2\pi/\log(T/2\pi)$ shrinks too slowly for fluctuations to stop exceeding ν before $T \sim 10^5$). The coefficients therefore depend on the cutoff — $(a, b) = (1.24, 1.42)$, $(1.71, 0.97)$, $(1.93, 0.83)$, $(2.07, 0.74)$ for ζ cut at $150, 320, 500, 811$ — and the table above compares ζ summed to 811 with characters summed to 150. The law must be *defined* at a common cutoff T_0; at $T_0 = 320$ the ζ fit is $\pm 2\%$ on all four scales with $(a, b) = (1.71, 0.97)$. The out-of-sample table is to be redone at a common T_0 once the characters are harvested to 320 (`code/harvest_weyl.py`); the seven fast characters, whose desert term dominates, are not expected to move much, but the exact figures await it (notebook §92).

What the desert alone explains. Computing the Slepian constant to arbitrary precision: $1 - \lambda_0(\tau\gamma_1) = 3.35 \times 10^{-6}$, 5.42×10^{-14}, 2.93×10^{-16} at $\mu = 3, 11, 16$, i.e. 75%, 28%, 21% of the measured depth. The union of all sub-Nyquist gaps gives 74 of 110 nats at $\mu = 11$ but is not a bound in either direction. The sharp constant is c_L itself.

Conjecture 1 (candidate form)**.** *The desert term is $-\ln\big(1 - \lambda_0(\frac{L}{2}\gamma_1)\big)$, Slepian's constant, in place of $a\,L(\gamma_1 - \nu)_+$. The gap term is the part not yet understood: the union concentration constant is not a bound in either direction, and the coefficient b is a fitted number. Whether a, b are absolute across L-functions is suggested by the out-of-sample table, not established.*

2 The quorum as a mechanism

Write $Q_S = Q + T_M$, T_M the towers of the missing primes M.

Lemma 2 (unconditional)**.** *Let Q, T_M be real symmetric matrices and v a unit vector. Put $w = T_M v$, $\kappa = \|w - (v^\top w)v\| > 0$, $u = (w - (v^\top w)v)/\kappa$, $a = v^\top Q_S v$, $d = u^\top Q_S u$, and $r = u^\top Q v$. If $a\,d < (\kappa + r)^2$ then Q_S is indefinite.*

Proof. The compression of Q_S to the orthonormal pair $\{v, u\}$ is $\begin{pmatrix} a & \kappa + r \\ \kappa + r & d \end{pmatrix}$. Its negative determinant gives one negative and one positive eigenvalue. Their Rayleigh quotients are also attained in the ambient space, so Q_S is indefinite. □

No eigenvector exactness and no hypothesis on zeros is used. Only for an *exact* eigenvector of Q does $r = 0$. For an approximate bottom vector, the residual must be enclosed: a sufficient condition is $ad < (\kappa - |r|)^2$ with $|r| < \kappa$. Taking v the exact bottom vector of Q gives $a = \varepsilon + \sum_{p \in M} \delta_p$, with $\varepsilon = \lambda_{\min}$ and $\delta_p = v^\top T_p v$ the *silence* of v to p; κ is its *coupling*. The v3 statement omitted r for arbitrary v and was false; the corrected statement supplies the residual condition needed by numerical witnesses and by lemmas B1/B2 of *lemma-quorum-scales*.

Certified instance, $\mu = 11$ (`code/quorum_2x2.py`, replay of 18 October 2026). Q and T_p are enclosed in Arb balls at 85 decimal digits, integration tolerance 10^{-58}, on the 47-dimensional cosine space V_{46}. The witness is the exact frozen dyadic vector from `witnesses_zeta_mu11_full.json`; it is not assumed to be an exact eigenvector. The implementation uses the unnormalized perpendicular vector and the full cross term, then divides the determinant by the product of the two squared norms. $T_{11} \equiv 0$ ($\Theta(L) = 0$), leaving voting primes $\{2, 3, 5, 7\}$. All fifteen nonempty missing-prime sets give strictly negative determinant enclosures. The closest to zero, $M = \{7\}$, is $-3.80631029173742 \times 10^{-9}$ (enclosure radius below 4.33×10^{-54} before decimal shortening). The complete enclosures, including the residual cross terms, are stored in `report/quorum-2x2-mu11.json`. This re-proves the fifteen $\mu = 11$ cases using one vector instead of their individual negative witnesses; it shares the full-form assembly and integration bounds with the existing certificate.

The μ-content (historical measurements). The following table records the earlier eigenvector-based measurements. Except for the corrected $\mu = 11$ replay above, its margins have not been re-certified with the residual cross term:

μ	ε	δ_p ($p = 2, 3, 5, 7, \ldots$)	κ_p	subsets	rel. margin
8	7.5×10^{-33}	0.052, 3.1e-4, 1.3e-10, 9.4e-21	0.70, 0.53, 3.6e-3, 8.8e-9	14/14	0.90
11	3.6×10^{-48}	0.058, 4.8e-4, 1.2e-9, 7.4e-17	0.74, 0.77, 0.051, 6.2e-5	14/14	0.85
16	1.9×10^{-73}	0.064, 6.6e-4, 4.9e-9, 4.0e-15, 2.0e-30, 1.2e-40	0.80, 0.88, 0.28, 6.7e-3, 1.5e-9, 2.5e-15	62/62	0.83

Two laws in p at fixed μ: $-\ln \delta_p \approx 7p - 12$ and $-\ln \kappa_p \approx 4p$ ($p \geq 5$). *The coupling decays at half the rate of the silence*; $\kappa_p^2/\delta_p \geq 9$ everywhere and grows with μ (for $p = 7$: $8 \times 10^3 \to 5 \times 10^7 \to 10^{10}$). These linear fits are local: with seven voting primes (χ_3, χ_4 at $\mu = 16, 22, 30$) both laws are read in the weight $w = p \log p$ — *silence* $-\ln \delta_p \approx 0.19\, s(\chi)\, w$ with one constant for ζ, χ_3, χ_4, and *coupling* $-\ln \kappa_p \approx c(\mu)\, w^2$, Gaussian, with $c(\mu) \approx 0.11\, s(\chi)/W$, $W = \mu \log \mu$ the weight of the window ($c = 0.029, 0.0068, 0.0045, 0.0051$ for ζ at 16, χ_3 at 22 and 30, χ_4 at 22; the coefficient decreases with μ). The margin of the single-vector certificate is then the parabola $\ln(\kappa_p^2/\delta_p) \approx 0.19\, s\, w\, (1 - w/W)$: positive for every prime inside the window, maximal at $w = W/2$, vanishing at the edge prime where $T_p \equiv 0$ — checked on 24 primes of four configurations to 10–20%. The bottom vector certifies the whole window at every μ measured; the quorum itself holds regardless. Measured; notebook §86–89. The worst case always contains 2 ($a \approx \delta_2 \approx 0.06$ against $\kappa^2 \approx 0.5$–0.9); two preregistered guesses died (the silence grows mildly with μ; the worst case is not the largest isolated prime).

What remains for every scale. Three unconditional analytic statements about the bottom vector, in the directions the lemma needs (silence small, coupling large): $\varepsilon(\mu) \le e^{-s\mu+b}$ (the desert, Section 1, there under RH), $\delta_p(\mu) \le e^{-7p+c}$, $\kappa_p(\mu) \ge e^{-4p-c'}$. What is measured is $\approx$; what is to be proved is $\le$ and $\ge$. The first is the depth law; the other two are new, concern the autocorrelation of an explicit eigenvector at the points $\log p$, and are where an analyst can enter.

Status

Lemma proved; $\mu = 11$ instance certified; everything else measured with stated numbers. Neither section moves RH: the first describes how small the positive floor on the line is, the second why removing a prime destroys it.